CSE203B Convex Optimization Homework 3

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Contributions:

Ethan did 3.13, 3.16, 2(d) and 3(e)
Gayathri did 3.36, 3.39, 2(c), 3(c), 3(d)
Tomasz did 3.1, 3.2, 2(a), 3(a), 3(b)
Tejal did 3.9, 3.14, 2(b), 2(e) and 3(f)

1 Exercises from Textbook

Solution for 3.1

Part (a): Proving the Convexity Inequality

Goal: Show that $f(x) \leq \frac{b-x}{b-a} f(a) + \frac{x-a}{b-a} f(b)$ for all $x \in [a, b]$.

Proof. By definition, a function $f : \mathbb{R} \to \mathbb{R}$ is convex if for any $\theta \in [0, 1]$:

$$f(\theta a + (1 - \theta)b) \leq \theta f(a) + (1 - \theta)f(b)$$

Let x be any point in the interval $[a, b]$. We can express x as a convex combination of the endpoints a and b. Let us define $\theta = \frac{b-x}{b-a}$. Since $a \leq x \leq b$, it follows that $0 \leq \theta \leq 1$.

The complementary weight is:

$$1 - \theta = 1 - \frac{b - x}{b - a} = \frac{b - a - (b - x)}{b - a} = \frac{x - a}{b - a}$$

We verify that this combination yields x:

$$\theta a + (1 - \theta)b = \frac{b - x}{b - a}a + \frac{x - a}{b - a}b = \frac{ab - ax + xb - ab}{b - a} = \frac{x(b - a)}{b - a} = x$$

Substitute these specific weights into the definition of convexity:

$$f(x) \leq \frac{b - x}{b - a} f(a) + \frac{x - a}{b - a} f(b)$$

□

Part (b): Proving the Slope Inequalities

Goal: Show $\frac{f(x)-f(a)}{x-a} \leq \frac{f(b)-f(a)}{b-a} \leq \frac{f(b)-f(x)}{b-x}$ for $x \in (a, b)$.

1. Proving the left inequality

We start with the result from Part (a). Subtract $f(a)$ from both sides:

$$\begin{aligned}
f(x) &\leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b) \\
f(x) - f(a) &\leq \left(\frac{b-x}{b-a} - 1\right) f(a) + \frac{x-a}{b-a}f(b) \\
f(x) - f(a) &\leq \left(\frac{b-x-(b-a)}{b-a}\right) f(a) + \frac{x-a}{b-a}f(b) \\
f(x) - f(a) &\leq \frac{a-x}{b-a}f(a) + \frac{x-a}{b-a}f(b) \\
f(x) - f(a) &\leq \frac{x-a}{b-a}\left(f(b) - f(a)\right)
\end{aligned}$$

Since $x > a$, we have $x - a > 0$. Dividing by $(x - a)$ preserves the inequality:

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(a)}{b - a}$$

2. Proving the right inequality

Start with the result from Part (a), but subtract $f(b)$ from both sides:

$$\begin{aligned}
f(x) - f(b) &\leq \frac{b-x}{b-a}f(a) + \left(\frac{x-a}{b-a} - 1\right) f(b) \\
f(x) - f(b) &\leq \frac{b-x}{b-a}f(a) + \frac{x-b}{b-a}f(b) \\
f(x) - f(b) &\leq \frac{b-x}{b-a}(f(a) - f(b))
\end{aligned}$$

Multiply by -1 (flipping the inequality):

$$f(b) - f(x) \geq \frac{b-x}{b-a}(f(b) - f(a))$$

Since $x < b$, we have $b - x > 0$. Divide by $(b - x)$:

$$\frac{f(b) - f(x)}{b - x} \geq \frac{f(b) - f(a)}{b - a} \implies \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(x)}{b - x}$$

Combining these yields the desired inequality.

Sketch Description

Geometrically, this represents the slopes of secant lines on a convex curve. Let points be $A = (a, f(a))$, $X = (x, f(x))$, and $B = (b, f(b))$.

- Slope of segment AX is $\frac{f(x)-f(a)}{x-a}$.
- Slope of segment AB is $\frac{f(b)-f(a)}{b-a}$.
- Slope of segment XB is $\frac{f(b)-f(x)}{b-x}$.

For a convex function, as we move from left to right, the slopes of the secant lines are non-decreasing: Slope$(AX) \leq$ Slope$(AB) \leq$ Slope(XB).

Part (c): Proving the Derivative Inequality

Goal: Show $f'(a) \leq \frac{f(b)-f(a)}{b-a} \leq f'(b)$.

Proof. Assume f is differentiable. We use the limits of the slopes found in Part (b).

Left side: From (b), for any $x \in (a, b)$, $\frac{f(x)-f(a)}{x-a} \leq \frac{f(b)-f(a)}{b-a}$. Taking the limit as $x \to a^+$:

$$\lim_{x \to a^+} \frac{f(x) - f(a)}{x - a} = f'(a) \implies f'(a) \leq \frac{f(b) - f(a)}{b - a}$$

Right side: From (b), $\frac{f(b)-f(a)}{b-a} \leq \frac{f(b)-f(x)}{b-x}$. Taking the limit as $x \to b^-$:

$$\lim_{x \to b^-} \frac{f(b) - f(x)}{b - x} = \lim_{x \to b^-} \frac{f(x) - f(b)}{x - b} = f'(b) \implies \frac{f(b) - f(a)}{b - a} \leq f'(b)$$

□

Part (d): Second Derivative Condition

Goal: Show $f''(a) \geq 0$ and $f''(b) \geq 0$.

Proof. Assume f is twice differentiable. From Part (c), we established that for any interval $[u, v]$ where $u < v$, $f'(u) \leq f'(v)$. This implies that f' is a non-decreasing function.

To show $f''(a) \geq 0$: For any $x > a$, applying the result from (c) to $[a, x]$ implies $f'(x) \geq f'(a)$.

$$f''(a) = \lim_{x \to a^+} \frac{f'(x) - f'(a)}{x - a}$$

Since $f'(x) - f'(a) \geq 0$ and $x - a > 0$, the limit is non-negative. Thus, $f''(a) \geq 0$.

To show $f''(b) \geq 0$: For any $x < b$, applying the result from (c) to $[x, b]$ implies $f'(x) \leq f'(b)$.

$$f''(b) = \lim_{x \to b^-} \frac{f'(b) - f'(x)}{b - x}$$

Since $f'(b) - f'(x) \geq 0$ and $b - x > 0$, the limit is non-negative. Thus, $f''(b) \geq 0$. □

Solution for 3.2

Analysis of the Top Figure

The top figure displays level sets formed by closed loops where the function values increase outwardly from 1 to 3, depicting a bowl shape. This function is both **quasiconvex** and **convex** because the sublevel sets—the regions enclosed by these curves—appear to be convex shapes without indentations. However, the function is **not quasiconcave** (and therefore **not concave**). The superlevel sets correspond to the regions *outside* the closed loops; these unbounded regions are inherently non-convex because they contain holes (the lower-valued inner regions).

Analysis of the Bottom Figure

In the bottom figure, the level curves are open lines that bow toward the right, with values increasing from left to right (from 1 to 6). This function is **not quasiconvex** (and therefore **not convex**). The sublevel sets, located to the left of the curves, are non-convex because the inward curvature of the contour lines creates a dented shape; a line segment connecting two points in this region could easily cross into the higher-valued region on the right. Conversely, the function appears to be **quasiconcave** and **concave** because the superlevel sets—the regions to the right of the curves—are convex shapes bounded by curves bowing away from them.

Solution for 3.9

Let $F \in \mathbb{R}^{n\times m}$ and $\hat{x} \in \mathbb{R}^n$. Define the restriction

$$\tilde{f}(z) = f(Fz + \hat{x}), \qquad \operatorname{dom} \tilde{f} = \{z \mid Fz + \hat{x} \in \operatorname{dom} f\}.$$

Assume f is twice differentiable and $\operatorname{dom} f$ is convex.

(a) We show that $\tilde{f}$ is convex on $\operatorname{dom} \tilde{f}$ iff

$$F^T \nabla^2 f(Fz + \hat{x}) F \succeq 0 \quad \forall z \in \operatorname{dom} \tilde{f}.$$

Since $\tilde{f}$ is a composition of f with an affine map, $\tilde{f}$ is twice differentiable on $\operatorname{dom} \tilde{f}$ and by the chain rule,

$$\nabla \tilde{f}(z) = F^T \nabla f(Fz + \hat{x}), \qquad \nabla^2 \tilde{f}(z) = F^T \nabla^2 f(Fz + \hat{x}) F.$$

Because $\operatorname{dom} \tilde{f}$ is convex (affine preimage of a convex set), the second-order characterization of convexity applies: a twice differentiable function on a convex domain is convex iff its Hessian is positive semidefinite everywhere. Hence

$$\tilde{f} \text{ convex on } \operatorname{dom} \tilde{f} \iff \nabla^2 \tilde{f}(z) \succeq 0 \ \forall z \in \operatorname{dom} \tilde{f} \iff F^T \nabla^2 f(Fz + \hat{x}) F \succeq 0 \ \forall z \in \operatorname{dom} \tilde{f},$$

as desired.

(b) Suppose $A \in \mathbb{R}^{p\times n}$ satisfies

$$AF = 0, \qquad \operatorname{rank} A = n - \operatorname{rank} F,$$

so that $\mathcal{N}(A) = \mathcal{R}(F)$ (nullspace equals the range of F). Show that $\tilde{f}$ is convex iff for all $z \in \operatorname{dom} \tilde{f}$ there exists $\lambda \in \mathbb{R}$ such that

$$\nabla^2 f(Fz + \hat{x}) + \lambda A^T A \succeq 0.$$

Fix $z \in \operatorname{dom} \tilde{f}$ and set

$$B := \nabla^2 f(Fz + \hat{x}) \in \mathbb{S}^n.$$

From part (a), $\tilde{f}$ is convex iff for all $z \in \operatorname{dom} \tilde{f}$,

$$F^T B F \succeq 0.$$

Equivalently, for all $w \in \mathbb{R}^m$,

$$w^T (F^T B F) w = (Fw)^T B (Fw) \ \geq \ 0.$$

Let $x := Fw$. Then $x \in \mathcal{R}(F) = \mathcal{N}(A)$, so the above is equivalent to

$$x^T B x \geq 0 \quad \forall x \in \mathcal{N}(A).$$

Now apply the given hint: if $B \in \mathbb{S}^n$ and $A \in \mathbb{R}^{p\times n}$, then

$$x^T B x \geq 0 \ \forall x \in \mathcal{N}(A) \quad \Longleftarrow \quad \exists \lambda \in \mathbb{R} \text{ s.t. } B + \lambda A^T A \succeq 0.$$

Thus, for each fixed z,

$$x^T \nabla^2 f(Fz + \hat{x}) x \geq 0 \ \forall x \in \mathcal{N}(A) \quad \Longleftarrow \quad \exists \lambda \in \mathbb{R} : \ \nabla^2 f(Fz + \hat{x}) + \lambda A^T A \succeq 0.$$

Since $\mathcal{N}(A) = \mathcal{R}(F)$, the left condition is exactly $F^T \nabla^2 f(Fz + \hat{x}) F \succeq 0$. Therefore, $\tilde{f}$ is convex if for all $z \in \operatorname{dom} \tilde{f}$ there exists λ with

$$\nabla^2 f(Fz + \hat{x}) + \lambda A^T A \succeq 0.$$

Combined with part (a), this yields the stated condition.

Solution 3.13

Using the hint,

$$D_{\mathrm{kl}} = f(u) - f(v) - \nabla f(v)^T(u-v), \quad f(v) = \sum_{i=1}^{n} v_i \log v_i$$

Proof of $D_{\mathrm{kl}}(u,v) \geq 0$

First, we check convexity of $f(v)$.

$$(\nabla f(v))_i = \frac{\partial}{\partial v_i}(v_i \log v_i) = \log v_i + 1$$
$$\nabla^2 f(v) = \operatorname{diag}\left(\frac{1}{v_1}, \frac{1}{v_2}, \cdots, \frac{1}{v_n}\right)$$

We are given $v \in \mathbf{R}^n_{++}$ so

$$\text{each } \frac{1}{v_i} > 0 \implies \nabla^2 f \succ 0$$

So f is **strictly convex**. Using the first-order condition,

$$\begin{aligned} f(u) &\geq f(v) + \nabla f(v)^T(u-v) \\ f(u) - f(v) - \nabla f(v)^T(u-v) &\geq 0 \\ \therefore D_{\mathrm{kl}}(u,v) &\geq 0 \end{aligned}$$

Proof of $D_{\mathrm{kl}}(u,v) = 0 \iff u = v$

Because f is strictly convex on $u, v \in \mathbf{R}^n_{++}$, we know the following inequality is strict when $u \neq v$

$$f(u) > f(v) + \nabla f(v)^T(u-v) \forall u \neq v$$

Therefore, $D_{\mathrm{kl}}(u,v) = 0$ holds true **only** if $u = v$.

Solution for 3.14

Let $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ with $\operatorname{dom} f = A \times B$ where $A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}^m$ are convex.

(a) Second-order condition. Assume f is twice differentiable. Write the block Hessian as

$$\nabla^2 f(x,z) = \begin{bmatrix} \nabla^2_{xx} f(x,z) & \nabla^2_{xz} f(x,z) \\ \nabla^2_{zx} f(x,z) & \nabla^2_{zz} f(x,z) \end{bmatrix}.$$

Then f is convex-concave (convex in x for each fixed z, concave in z for each fixed x) iff for all $(x,z) \in A \times B$,

$$\nabla^2_{xx} f(x,z) \succeq 0 \qquad \text{and} \qquad \nabla^2_{zz} f(x,z) \preceq 0.$$

There is no sign restriction on the mixed blocks $\nabla^2_{xz} f$ and $\nabla^2_{zx} f$.

(b) Saddle-point property and strong max–min. Assume f is convex-concave and differentiable, and that

$$\nabla f(\tilde{x}, \tilde{z}) = 0, \quad \text{i.e.} \quad \nabla_x f(\tilde{x}, \tilde{z}) = 0, \quad \nabla_z f(\tilde{x}, \tilde{z}) = 0.$$

Fix $\tilde{z}$. Since $x \mapsto f(x, \tilde{z})$ is convex on A, the first-order condition for convexity gives

$$f(x, \tilde{z}) \geq f(\tilde{x}, \tilde{z}) + \nabla_x f(\tilde{x}, \tilde{z})^T (x - \tilde{x}) = f(\tilde{x}, \tilde{z}) \qquad \forall x \in A.$$

Thus $f(\tilde{x}, \tilde{z}) \leq f(x, \tilde{z})$ for all x.

Fix $\tilde{x}$. Since $z \mapsto f(\tilde{x}, z)$ is concave on B, the first-order condition for concavity gives

$$f(\tilde{x}, z) \leq f(\tilde{x}, \tilde{z}) + \nabla_z f(\tilde{x}, \tilde{z})^T (z - \tilde{z}) = f(\tilde{x}, \tilde{z}) \qquad \forall z \in B.$$

Thus $f(\tilde{x}, z) \leq f(\tilde{x}, \tilde{z})$ for all z.

Combining the two inequalities yields the saddle-point property:

$$f(\tilde{x}, z) \leq f(\tilde{x}, \tilde{z}) \leq f(x, \tilde{z}) \qquad \forall x \in A, \ \forall z \in B.$$

Now show the strong max–min property. From $f(\tilde{x}, z) \leq f(\tilde{x}, \tilde{z})$ for all z,

$$\sup_{z \in B} f(\tilde{x}, z) = f(\tilde{x}, \tilde{z}).$$

From $f(\tilde{x}, \tilde{z}) \leq f(x, \tilde{z})$ for all x,

$$\inf_{x \in A} f(x, \tilde{z}) = f(\tilde{x}, \tilde{z}).$$

Also, for any $z \in B$, since $\inf_x f(x, z) \leq f(\tilde{x}, z) \leq f(\tilde{x}, \tilde{z})$,

$$\sup_{z \in B} \inf_{x \in A} f(x, z) \leq f(\tilde{x}, \tilde{z}).$$

For any $x \in A$, since $\sup_z f(x, z) \geq f(x, \tilde{z}) \geq f(\tilde{x}, \tilde{z})$,

$$\inf_{x \in A} \sup_{z \in B} f(x, z) \geq f(\tilde{x}, \tilde{z}).$$

But in general we always have $\sup_z \inf_x f(x, z) \leq \inf_x \sup_z f(x, z)$, hence both sides must equal $f(\tilde{x}, \tilde{z})$:

$$\sup_{z \in B} \inf_{x \in A} f(x, z) = \inf_{x \in A} \sup_{z \in B} f(x, z) = f(\tilde{x}, \tilde{z}).$$

(c) Saddle-point property implies stationarity. Assume f is differentiable and for some $(\tilde{x}, \tilde{z}) \in A \times B$,

$$f(\tilde{x}, z) \leq f(\tilde{x}, \tilde{z}) \leq f(x, \tilde{z}) \qquad \forall x \in A, \ \forall z \in B.$$

Since A is convex and $\tilde{x}$ minimizes $x \mapsto f(x, \tilde{z})$ over A, we have

$$f(\tilde{x}, \tilde{z}) \leq f(x, \tilde{z}) \quad \forall x \in A.$$

If $\tilde{x}$ lies in the interior of A, then every direction is feasible, and the first-order optimality condition gives

$$\nabla_x f(\tilde{x}, \tilde{z}) = 0.$$

In particular, if $A = \mathbb{R}^n$, then $\tilde{x}$ is an interior point and hence $\nabla_x f(\tilde{x}, \tilde{z}) = 0$.

Similarly, since $\tilde{z}$ maximizes $z \mapsto f(\tilde{x}, z)$ over B, we have

$$f(\tilde{x}, z) \leq f(\tilde{x}, \tilde{z}) \quad \forall z \in B.$$

If $\tilde{z}$ lies in the interior of B, then the first-order optimality condition for maximization implies

$$\nabla_z f(\tilde{x}, \tilde{z}) = 0.$$

In particular, if $B = \mathbb{R}^m$, then $\nabla_z f(\tilde{x}, \tilde{z}) = 0$. Therefore, when $A = \mathbb{R}^n$ and $B = \mathbb{R}^m$, we obtain

$$\nabla f(\tilde{x}, \tilde{z}) = 0.$$

Solution 3.16

(a) $f(x) = e^x - 1$ on $\mathbf{R}$

$$\nabla^2 f(x) = e^x > 0 \quad \forall x, \quad \boxed{\text{convex}} \text{ and } \boxed{\text{quasiconvex}}$$

(b) $f(x_1, x_2) = x_1 x_2$ on $\mathbf{R}^2_{++}$

$$\nabla^2 f = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$
$$\det(\nabla^2 f) = 0^2 - 1^2 = -1$$

A negative det means our Hessian has eigenvalues of opposite signs. A function is convex only if Hessian is positive semidefinite (PSD), so all $\lambda \geq 0$ and its concave only if Hessian is negative semidefinite, so all $\lambda \leq 0$. Neither of these are true; the Hessian is indefinite.

Proof of quasiconcavity:

For $t > 0$,

$$S_t = \{(x_1, x_2) : x_1 x_2 \geq t\} = \{x : \log(x_1 x_2) \geq \log t\} = \{x : g(x) \geq \log t\}$$

We make the claim that if g is concave, then $\{x : g(x) \geq c\}$ is convex.

Proof: For any x, y in set and $\theta \in [0, 1]$, we know from the concavity property that

$$g(\theta x + (1-\theta)y) \geq \theta g(x) + (1-\theta) g(y) \geq \theta c (1-\theta) c = c$$

So $\theta x + (1-\theta)y$ is also in the set. Thus, S_t is convex for all t, so f is $\boxed{\text{quasiconcave}}$.

We could prove it more simply:

Define

$$S_t = \{(x_1, x_2) : x_1 x_2 \geq t\}$$

If $t \leq 0$, then $S_t = \mathbf{R}^2_{++}$ = convex.
If $t > 0$,

$$x_1 x_2 \geq t \iff \log x_1 + \log x_2 \geq \log t$$

log is concave implies that $\log x_1 + \log x_2$ is concave. The set where a concave function is $\geq$ a constant is convex. Thus, S_t is convex, so f is $\boxed{\text{quasiconcave}}$.

(c) $f(x_1, x_2) = 1/(x_1 x_2)$ on $\mathbf{R}^2_{++}$

$$\nabla^2 f(x_1, x_2) = \frac{1}{x_1^2 x_2^2} \begin{bmatrix} \frac{2x_2}{x_1} & 1 \\ 1 & \frac{2x_1}{x_2} \end{bmatrix}$$

This Hessian is positive definite on $\mathbf{R}^2_{++}$
so f is $\boxed{\text{convex}}$, and thus $\boxed{\text{quasiconvex}}$.

It is not concave (needs negative semidefinite).

For $t > 0$, the following is not convex

$$1/(x_1 x_2) \geq t\} \iff \{x_1 x_2 \leq 1/t\}$$

so f is not quasiconcave.

(d) $f(x_1, x_2) = x_1/x_2$ on $\mathbf{R}^2_{++}$

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} 0 & -\frac{1}{x_2^2} \\ -\frac{1}{x_2^2} & 2\frac{x_1}{x_2^3} \end{bmatrix}$$
$$\det(\nabla^2 f) = 0 - \left(-\frac{1}{x_2^2}\right)^2 = -\frac{1}{x_2^4} < 0$$

Hessian determinant < 0 so f is neither convex nor concave.

Now, we can check for quasiconvexity and quasiconcavity.

For quasiconvexity, we want to see that all sublevel sets S_t are convex for all $t \in \mathbb{R}$

$$S_t = \{x \in \mathrm{dom} f \mid f(x) \leq t\}$$

We can rewrite the condition

$$\frac{x_1}{x_2} \leq t \implies x_1 \leq t x_2 \implies x_1 - t x_2 \leq 0$$

where we use the fact that x_2 is non-zero so we can multiply by it on both sides.

The above is a linear inequality; it defines a half-plane as it intersects with the first quadrant $\mathbf{R}^2_{++}$. Half-planes are convex; intersection of two convex sets is convex, so S_t is convex. $\therefore f$ is $\boxed{\text{quasiconvex}}$.

We can do something very similar for quasiconcavity.

Let superlevel sets $S_t = \{x \in \mathrm{dom} f \mid f(x) \geq t\}$

With the same logic, we get $x_1 - t x_2 \geq 0$. This is also a linear inequality and defines a half-plane. The intersection of a halfplane with the convex domain/set $\mathbf{R}^2_{++}$ is convex $\therefore f$ is $\boxed{\text{quasiconcave}}$.

(e) $f(x_1, x_2) = x_1^2/x_2$ on $\mathbf{R} \times \mathbf{R}_{++}$

This is a quadratic-over-linear function. We know

$$\text{If } x_2 > 0, \text{ then } f(x_1, x_2) = \frac{x_1^2}{x_2^2} \text{ is convex}$$

We can use epigraphs, defined as

$$\mathrm{epi}(f) = \left\{(x_1, x_2, t) : x_2 > 0, t \geq \frac{x_1^2}{x_2}\right\}$$

Rewriting, we get

$$t \geq \frac{x_1^2}{x_2} \iff x_1^2 \leq t x_2 \iff \begin{bmatrix} t & x_1 \\ x_1 & x_2 \end{bmatrix} \succeq 0$$

We have a linear matrix inequality, which defines a convex set, making the epigraph convex $\therefore f$ is $\boxed{\text{convex}}$ and $\boxed{\text{quasiconvex}}$.

Now, let's check concavity

Let

$$S_\alpha = \{(x_1, x_2) : x_2 > 0, x_1^2 \geq \alpha x_2\}, \quad \alpha > 0$$

We can pick two points in S_α

$$x = (\sqrt{\alpha}, 1), \quad y = (-\sqrt{\alpha}, 1)$$

$$x_1^2 = \alpha = \alpha x_2, \quad y_1^2 = \alpha = \alpha x_2$$

Midpoint is

$$\frac{x+y}{2} = (0, 1)$$

$0^2 = 0 \ngeq \alpha \cdot 1$ so the midpoint is not in S_α. Therefore, S_α is not convex and f is not quasiconcave, thus not concave either.

(f) $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, where $0 \leq \alpha \leq 1$, on $\mathbf{R}^2_{++}$

$$\begin{aligned}
\frac{\partial f}{\partial x_1} &= \alpha x_1^{\alpha-1} x_2^{1-\alpha} = \frac{\alpha f}{x_1} \\
\frac{\partial f}{\partial x_2} &= (1-\alpha) x_1^\alpha x_2^{-\alpha} = \frac{(1-\alpha) f}{x_2} \\
\nabla^2 f &= \begin{bmatrix} \frac{\alpha(\alpha-1)f}{x_1^2} & \frac{\alpha(1-\alpha)f}{x_1 x_2} \\ \frac{\alpha(1-\alpha)f}{x_1 x_2} & -\frac{\alpha(1-\alpha)f}{x_2^2} \end{bmatrix}
\end{aligned}$$

For $0 < \alpha < 1$, we can simplify the Hessian above to

$$\nabla^2 f = -\alpha(1-\alpha) f \begin{bmatrix} \frac{1}{x_1^2} & -\frac{1}{x_1 x_2} \\ -\frac{1}{x_1 x_2} & \frac{1}{x_2^2} \end{bmatrix}$$

Now we can show let's check the matrix (without the coefficients) is PSD. Let's call that matrix A

$$y^T A y = \frac{y_1^2}{x_1^2} - 2\frac{y_1 y_2}{x_1 x_2} + \frac{y_2^2}{x_2^2} = \left(\frac{y_1}{x_1} - \frac{y_2}{x_2}\right)^2 \geq 0$$

So A is PSD. Since $\alpha(1-\alpha)f > 0$ on $\mathbf{R}^2_{++}$, we know $-\alpha(1-\alpha)f < 0$, so $\nabla^2 f \preceq 0$

$\therefore f$ is $\boxed{\text{concave}}$, and thus also $\boxed{\text{quasiconcave}}$, for $0 \leq alpha \leq 1$

We can show that $f = x_2$ when $\alpha = 0$ and $f = x_1$ when $\alpha = 1$. Both these are affine, so f is also concave at the endpoints.

Solution for 3.36

Part (a)

The function is $f(x) = \max_{i=1,\ldots,n} x_i$. The conjugate is defined as:

$$f^*(y) = \sup_x (y^T x - \max_i x_i)$$

If $y \succeq 0$ and $\mathbf{1}^T y = 1$, then $y^T x$ is a convex combination of x_i, which implies $y^T x \leq \max_i x_i$. The maximum value of $y^T x - \max_i x_i$ is 0. If these conditions on y are not met, the supremum is ∞.

$$f^*(y) = \begin{cases} 0 & y \succeq 0, \mathbf{1}^T y = 1 \\ \infty & \text{otherwise} \end{cases}$$

Part (b)

The function $f(x) = \sum_{i=1}^{r} x_{[i]}$ can be given as:

$$f(x) = \sup\{z^T x \mid 0 \preceq z \preceq \mathbf{1}, \mathbf{1}^T z = r\}$$

The conjugate is defined as $f^*(y) = \sup_x(y^T x - f(x))$. Substituting the dual representation:

$$f^*(y) = \sup_x \left(y^T x - \sup_{z \in \mathcal{Z}} z^T x \right)$$

where $\mathcal{Z} = \{z \in \mathbb{R}^n \mid 0 \leq z_i \leq 1, \sum z_i = r\}$.
If $y \in \mathcal{Z}$, then $y^T x \leq \sup_{z \in \mathcal{Z}} z^T x$ for all x, and the supremum is attained at 0. If $y \notin \mathcal{Z}$, we can choose x such that the expression is unbounded. Thus:

$$f^*(y) = \begin{cases} 0 & 0 \preceq y \preceq \mathbf{1}, \mathbf{1}^T y = r \\ \infty & \text{otherwise} \end{cases}$$

Part(c)

Let $f(x) = \max_{i=1,\ldots,m}(a_i x + b_i)$ with $a_1 < a_2 < \cdots < a_m$. The conjugate is:

$$f^*(y) = \sup_x \left(yx - \max_i (a_i x + b_i) \right)$$

The supremum is finite iff $y \in [a_1, a_m]$. For $y \in [a_i, a_{i+1}]$, the supremum occurs at the intersection where $a_i x + b_i = a_{i+1} x + b_{i+1}$. This results in a piecewise linear convex function:

$$f^*(y) = -b_i \quad \text{for } y = a_i$$

More generally, for $y \in [a_1, a_m]$, it is the linear interpolation:

$$f^*(y) = \min\{\sum \theta_i(-b_i) \mid \sum \theta_i a_i = y, \sum \theta_i = 1, \theta_i \geq 0\}$$

Part (d)

For $f(x) = x^p$ on $\mathbb{R}_{++}$:

1. **Case I** $p > 1$**:** The function is convex. Setting $\nabla_x(yx - x^p) = 0$ gives $x = (y/p)^{1/(p-1)}$.

$$f^*(y) = (p-1)\left(\frac{y}{p}\right)^{p/(p-1)} \quad \text{for } y \geq 0$$

   $f^*(y) = \frac{1}{q}(y/p)^q$ where $1/p + 1/q = 1$.

2. **Case II** $p < 0$**:** The function is also convex on $\mathbb{R}_{++}$. The supremum is finite only for $y \leq 0$.

$$f^*(y) = -(-y/p)^{q/p}(1-p) \quad \text{for } y \leq 0$$

Part(e)

The function is $f(x) = -(\prod_{i=1}^{n} x_i)^{1/n}$ with dom $f = \mathbb{R}^n_{++}$. The conjugate is:

$$f^*(y) = \sup_{x \succ 0} \left(y^T x + \left(\prod_{i=1}^{n} x_i \right)^{1/n} \right)$$

If any $y_i > 0$, the expression is unbounded as $x_i \to \infty$. Thus, we must have $y \preceq 0$.

Since $f(x)$ is positively homogeneous of degree 1, $f^*(y)$ is the indicator function of a set. Specifically, the supremum is 0 if $y^T x - f(x) \leq 0$ for all x, and ∞ otherwise.

Using the AM-GM inequality or stationary point analysis, the condition for the supremum to be 0 is that the geometric mean of the components of $-y$ is at least $1/n$:

$$f^*(y) = \begin{cases} 0 & y \preceq 0, \ \left(\prod_{i=1}^n (-y_i)\right)^{1/n} \geq 1/n \\ \infty & \text{otherwise} \end{cases}$$

Part (f)

$f(x,t) = -\log(t^2 - x^T x)$ with domain $\{(x,t) \in \mathbb{R}^n \times \mathbb{R} \mid \|x\|_2 < t\}$. The conjugate is:

$$f^*(y,v) = \sup_{\|x\|_2 < t} \left(y^T x + vt + \log(t^2 - x^T x)\right)$$

Stationary Points

Setting the gradients with respect to x and t to zero:

$$\nabla_x : y = \frac{2x}{t^2 - x^T x}, \quad \nabla_t : v = -\frac{2t}{t^2 - x^T x}$$

Note that for a solution to exist, we must have $v < 0$ and $v^2 > \|y\|_2^2$ (the dual variables must lie in the negative dual cone).

Solving the System

From the ratio of the gradients, we find $x = -(t/v)y$. Substituting this into the expression for v:

$$v(t^2 - (-t/v)^2 \|y\|_2^2) = -2t \implies t(v^2 - \|y\|_2^2) = -2v$$

This yields the optimal values:

$$t^* = \frac{-2v}{v^2 - \|y\|_2^2}, \quad x^* = \frac{2y}{v^2 - \|y\|_2^2}$$

Result

Substituting (x^*, t^*) back into the conjugate definition:

$$f^*(y,v) = y^T x^* + vt^* + \log((t^*)^2 - \|x^*\|_2^2)$$

Simplifying the terms, we obtain:

$$f^*(y,v) = \begin{cases} -2 + 2\log 2 - \log(v^2 - y^T y) & \|y\|_2 < -v \\ \infty & \text{otherwise} \end{cases}$$

The domain is written as $v > \|y\|_2$ by changing the sign of the dual variable, but based on the supremum definition, v must be negative here.

Solution for 3.40

Part (a): Gradient of the Conjugate Function

The conjugate function is defined as:

$$f^*(y) = \sup_{x \in \text{dom } f} (y^T x - f(x))$$

For a given y, let $x(y)$ be the point that achieves the supremum. Since f is convex and differentiable, the supremum is attained when the gradient of the objective $g(x) = y^T x - f(x)$ with respect to x is zero:

$$\nabla_x(y^T x - f(x)) = y - \nabla f(x) = 0 \implies y = \nabla f(x)$$

Given that $\bar{y} = \nabla f(\bar{x})$ and $\nabla^2 f(\bar{x}) \succ 0$, f is locally strictly convex, making $\bar{x}$ the unique maximizer for $y = \bar{y}$. Thus:

$$f^*(\bar{y}) = \bar{y}^T \bar{x} - f(\bar{x})$$

To find the gradient $\nabla f^*(\bar{y})$, we differentiate $f^*(y) = y^T x(y) - f(x(y))$ with respect to y using the chain rule:

$$\nabla f^*(y) = x(y) + (\nabla_y x(y))^T y - (\nabla_y x(y))^T \nabla f(x(y))$$

$$\nabla f^*(y) = x(y) + (\nabla_y x(y))^T (y - \nabla f(x(y)))$$

From the optimality condition $y = \nabla f(x(y))$, the term $(y - \nabla f(x(y)))$ is zero. Therefore:

$$\nabla f^*(\bar{y}) = \bar{x}$$

Part (b): Hessian of the Conjugate Function

From part (a), we established the identity:

$$\nabla f^*(y) = x$$

where x and y are related by the gradient mapping $y = \nabla f(x)$.

To find the Hessian $\nabla^2 f^*(y)$, we differentiate the identity $\nabla f^*(y) = x(y)$ with respect to y:

$$\nabla^2 f^*(y) = \frac{dx}{dy}$$

We can find $\frac{dx}{dy}$ by applying implicit differentiation to the relation $y = \nabla f(x)$. Differentiating both sides with respect to y:

$$\nabla_y(y) = \nabla_y(\nabla f(x))$$

$$I = \nabla^2 f(x) \frac{dx}{dy}$$

Substituting $\frac{dx}{dy} = \nabla^2 f^*(y)$ into the equation:

$$I = \nabla^2 f(x) \nabla^2 f^*(y)$$

Since $\nabla^2 f(\bar{x}) \succ 0$, the Hessian matrix is non-singular and invertible. Multiplying both sides by $\nabla^2 f(\bar{x})^{-1}$ gives:

$$\nabla^2 f^*(\bar{y}) = \left(\nabla^2 f(\bar{x})\right)^{-1}$$

2 Convex Functions

Solution for (a)

A twice-differentiable function is convex on its domain if its second derivative is non-negative ($f''(x) \geq 0$) for all points in the domain.

Let $f(x) = e^{e^x} - \log(x)$. (Assuming $\log(x)$ denotes the natural logarithm, $\ln(x)$).

Applying the chain rule to the first term and the standard derivative rule to the logarithmic term.

$$f'(x) = \frac{d}{dx}\left(e^{e^x}\right) - \frac{d}{dx}(\log(x))$$

Using the chain rule on e^{e^x}: Let $u = e^x$, then $\frac{d}{dx}e^u = e^u \cdot u'$.

$$\frac{d}{dx}\left(e^{e^x}\right) = e^{e^x} \cdot e^x$$

Thus, the first derivative is:

$$f'(x) = e^{e^x} \cdot e^x - \frac{1}{x}$$

Next we differentiate $f'(x)$ to find $f''(x)$.

$$f''(x) = \frac{d}{dx}\left(e^{e^x} \cdot e^x\right) - \frac{d}{dx}\left(x^{-1}\right)$$

Using the product rule for the first term:

$$\frac{d}{dx}\left(e^{e^x} \cdot e^x\right) = \left(\frac{d}{dx}e^{e^x}\right) \cdot e^x + e^{e^x} \cdot \left(\frac{d}{dx}e^x\right)$$

Substitute the result from Step 1 ($\frac{d}{dx}e^{e^x} = e^{e^x}e^x$):

$$= (e^{e^x} \cdot e^x) \cdot e^x + e^{e^x} \cdot e^x$$

$$= e^{e^x} \cdot e^{2x} + e^{e^x} \cdot e^x$$

$$= e^{e^x}(e^{2x} + e^x)$$

Differentiating the second term:

$$\frac{d}{dx}\left(-x^{-1}\right) = -(-1)x^{-2} = \frac{1}{x^2}$$

Combining these, the second derivative is:

$$f''(x) = e^{e^x}(e^{2x} + e^x) + \frac{1}{x^2}$$

The domain is $\mathbb{R}_{++}$, meaning $x > 0$. We analyze the sign of each term in $f''(x)$:

The exponential function e^y is always positive for any real number y. Therefore, $e^{e^x} > 0$, $e^{2x} > 0$, and $e^x > 0$.

The product $e^{e^x}(e^{2x} + e^x)$ is the product of positive terms, so it is strictly positive.

Since $x > 0$, it follows that $x^2 > 0$, and thus $\frac{1}{x^2} > 0$.

Since $f''(x)$ is the sum of two strictly positive terms:

$$f''(x) > 0 \quad \text{for all } x \in \mathbb{R}_{++}$$

Since $f''(x) \geq 0$ for all x in the domain, the function $f(x)$ is **convex**.

Solution for (b)

$$f(x) = x^2 \log x, \quad \operatorname{dom} f = \mathbb{R}_+.$$

We use the second-derivative test on $[0, \infty)$. First,

$$f'(x) = \frac{d}{dx}\big(x^2 \log x\big) = 2x \log x + x.$$

Then,

$$f''(x) = \frac{d}{dx}\big(2x \log x + x\big) = 2 \log x + 3.$$

Hence $f''(x) \geq 0$ iff $2 \log x + 3 \geq 0$, i.e.,

$$\log x \geq -\frac{3}{2} \quad \Longleftrightarrow \quad x \geq e^{-3/2}.$$

Therefore, f is **convex** on $[e^{-3/2}, \infty)$ and **concave** on $[0, e^{-3/2})$. In particular, f is ***not* convex** on all of $\mathbb{R}_+$.

Solution for (c)

$f(x) = \min\{x_1, x_2^2, \ldots, x_n^n\}$ The function $f(x) = \min\{x_1, x_2^2, \ldots, x_n^n\}$ is **neither convex nor concave** over the domain $\mathbb{R}^n$.

Proof of Non-concavity: A function is concave if $f(\theta u + (1-\theta)v) \geq \theta f(u) + (1-\theta) f(v)$ for all $u, v \in \operatorname{dom} f$ and $\theta \in [0, 1]$. Let $n = 2$, $u = (10, 1)$, and $v = (10, -1)$.

$$f(u) = \min\{10, 1^2\} = 1, \quad f(v) = \min\{10, (-1)^2\} = 1$$

For $\theta = 0.5$, the midpoint is $w = (10, 0)$:

$$f(w) = \min\{10, 0^2\} = 0$$

Since $0 < 0.5(1) + 0.5(1)$, the constraint for concavity is violated.

Proof of Non-convexity: A function is convex if $f(\theta u + (1-\theta)v) \leq \theta f(u) + (1-\theta) f(v)$. Let $n = 2$, $u = (0, 10)$, and $v = (10, 0)$.

$$f(u) = \min\{0, 100\} = 0, \quad f(v) = \min\{10, 0\} = 0$$

For $\theta = 0.5$, the midpoint is $w = (5, 5)$:

$$f(w) = \min\{5, 25\} = 5$$

Since $5 > 0.5(0) + 0.5(0)$, the Jensen inequality for convexity is violated.

Furthermore, for odd $i > 1$, the individual components x_i^i are themselves neither convex nor concave on $\mathbb{R}$. Thus, the function is neither convex nor concave.

Solution for (d)

Let

$$g(t) = f(X + tV) = \operatorname{tr}((X + tV)^{-1})$$

We can get

$$X + tV = X^{1/2}(I + tX^{-1/2} V X^{-1/2}) X^{1/2}$$

Let $A = X^{-1/2} V X^{-1/2}$. Since V is symmetric, we know A is also symmetric.

$$(X+tV)^{-1} = X^{-1/2}(I+tA)^{-1}X^{-1/2}$$

Then using $\mathrm{tr}(AB) = \mathrm{tr}(BA)$,

$$\begin{aligned} g(t) &= \mathrm{tr}(X^{-1/2}(I+tA)^{-1}X^{-1/2}) = \mathrm{tr}(X^{-1}(I+tA)^{-1}) \\ &= \sum_{i=1}^{n} \frac{c_i}{1+t\lambda_i} \end{aligned}$$

$$g''(t) = \sum_{i=1}^{n} \frac{2c_i\lambda_i^2}{(1+t\lambda_i)^3}$$

When $t=0$, $g''(0) = \sum_{i=1}^{n} 2c_i\lambda_i^2 \geq 0$. We can say that this inequality holds for $X+tV \in \mathbb{S}^n_{++}$ $\therefore f$ is $\boxed{\text{convex}}$ on $\mathrm{dom} f$.

$f(x,y) = \dfrac{1}{xy}, \quad \mathrm{dom} f = \mathbb{R}^2_{++}$.

Solution for (e)

$$f(x,y) = \frac{1}{xy}, \quad \mathrm{dom}\, f = \mathbb{R}^2_{++}.$$

Compute the Hessian. First partial derivatives:

$$f_x(x,y) = -\frac{1}{x^2y}, \qquad f_y(x,y) = -\frac{1}{xy^2}.$$

Second partial derivatives:

$$f_{xx}(x,y) = \frac{2}{x^3y}, \qquad f_{yy}(x,y) = \frac{2}{xy^3}, \qquad f_{xy}(x,y) = \frac{1}{x^2y^2}.$$

Thus the Hessian is

$$\nabla^2 f(x,y) = \begin{bmatrix} \frac{2}{x^3y} & \frac{1}{x^2y^2} \\ \frac{1}{x^2y^2} & \frac{2}{xy^3} \end{bmatrix}.$$

For $(x,y) \in \mathbb{R}^2_{++}$, the leading principal minor is

$$\frac{2}{x^3y} > 0,$$

and the determinant is

$$\det(\nabla^2 f) = \frac{2}{x^3y} \cdot \frac{2}{xy^3} - \left(\frac{1}{x^2y^2}\right)^2 = \frac{4}{x^4y^4} - \frac{1}{x^4y^4} = \frac{3}{x^4y^4} > 0.$$

Hence $\nabla^2 f(x,y)$ is positive definite for all $(x,y) \in \mathbb{R}^2_{++}$, so f is **strictly convex** on $\mathbb{R}^2_{++}$.

3 Conjugate Function

Solution for (a)

$$f(x) = e^{x+1}$$

We evaluate the conjugate function defined by:

$$f^*(y) = \sup_{x\in\mathbb{R}} \left(yx - e^{x+1}\right)$$

Let $g(x) = yx - e^{x+1}$. To find the supremum, we find the critical points by setting the derivative with respect to x to zero:

$$\frac{dg}{dx} = y - e^{x+1} = 0 \implies e^{x+1} = y$$

This equation has a real solution for x only if $y > 0$.

- **Case 1:** $y > 0$
  Solving for x:
  $$x + 1 = \ln(y) \implies x^* = \ln(y) - 1$$
  Since $g''(x) = -e^{x+1} < 0$, this critical point is a maximum. Substituting x^* back into the expression for $f^*(y)$:
  $$\begin{aligned} f^*(y) &= y(\ln(y) - 1) - e^{(\ln(y)-1)+1} \\ &= y\ln(y) - y - e^{\ln(y)} \\ &= y\ln(y) - y - y \\ &= y\ln(y) - 2y \end{aligned}$$

- **Case 2:** $y = 0$
  The expression becomes $\sup_x(-e^{x+1})$. As $x \to -\infty$, $-e^{x+1} \to 0$. Thus, $f^*(0) = 0$.

- **Case 3:** $y < 0$
  Let $x \to -\infty$. Then $yx \to +\infty$ and $e^{x+1} \to 0$. The expression approaches $+\infty$.

Result:

$$f^*(y) = \begin{cases} y\ln(y) - 2y & \text{if } y \geq 0 \\ \infty & \text{if } y < 0 \end{cases}$$

Solution for (b)

$$f(x) = ax^3 + b$$

We evaluate:

$$f^*(y) = \sup_{x\in\mathbb{R}} \left(yx - (ax^3 + b)\right) = \sup_{x\in\mathbb{R}} \left(yx - ax^3 - b\right)$$

We analyze this based on the parameter a.

- **Case 1:** $a \neq 0$
  The function $f(x)$ is a cubic polynomial. A cubic function on an unbounded domain cannot be dominated by a linear function yx.
  - If $a > 0$: Consider $x \to -\infty$. Then x^3 is large negative, so $-ax^3$ is large positive. The term dominates yx, so the expression goes to $+\infty$.
  - If $a < 0$: Consider $x \to +\infty$. Then x^3 is large positive, so $-ax^3$ is large positive (since $-a > 0$). The expression goes to $+\infty$.

  In both instances, the supremum is unbounded.
  $$f^*(y) = \infty, \quad \forall y \in \mathbb{R}$$

- **Case 2:** $a = 0$
  The function becomes $f(x) = b$ (a constant).
  $$f^*(y) = \sup_{x\in\mathbb{R}}(yx - b)$$
  This is the conjugate of a constant function.

- If $y = 0$, the expression is $-b$.
- If $y \neq 0$, we can choose x (with the same sign as y) to make yx arbitrarily large, so the supremum is $+\infty$.

Result:

- If $a \neq 0$:
$$f^*(y) = \infty$$
- If $a = 0$:
$$f^*(y) = \begin{cases} -b & \text{if } y = 0 \\ \infty & \text{if } y \neq 0 \end{cases}$$

Solution for (c)

Let $f(x) = x^T Q x + a^T x$ where $Q \in \mathbb{S}^n_{++}$. The conjugate function is defined as:

$$f^*(y) = \sup_x \left(y^T x - x^T Q x - a^T x \right)$$

To find the supremum, we take the gradient with respect to x and set it to zero:

$$\nabla_x (y^T x - x^T Q x - a^T x) = y - 2Qx - a = 0$$

Solving for x:

$$2Qx = y - a \implies x^* = \frac{1}{2} Q^{-1}(y - a)$$

Substituting x^* back into the expression for the conjugate:

$$f^*(y) = y^T \left(\frac{1}{2} Q^{-1}(y-a) \right) - \left(\frac{1}{2} Q^{-1}(y-a) \right)^T Q \left(\frac{1}{2} Q^{-1}(y-a) \right) - a^T \left(\frac{1}{2} Q^{-1}(y-a) \right)$$

Factoring out terms and using the symmetry of Q^{-1}:

$$f^*(y) = \frac{1}{2}(y-a)^T Q^{-1}(y-a) + \frac{1}{2} a^T Q^{-1}(y-a) - \frac{1}{4}(y-a)^T Q^{-1}(y-a) - \frac{1}{2} a^T Q^{-1}(y-a)$$

$$f^*(y) = \frac{1}{4}(y-a)^T Q^{-1}(y-a)$$

Solution for (d)

Let $f(x) = \frac{1}{2}\|x\|_2^2$. The conjugate is defined as:

$$f^*(y) = \sup_x \left(y^T x - \frac{1}{2} x^T x \right)$$

Setting the gradient with respect to x to zero:

$$\nabla_x (y^T x - \frac{1}{2} x^T x) = y - x = 0 \implies x^* = y$$

Substituting $x^* = y$ back into the objective:

$$f^*(y) = y^T(y) - \frac{1}{2} y^T y = \|y\|_2^2 - \frac{1}{2}\|y\|_2^2$$

$$f^*(y) = \frac{1}{2}\|y\|_2^2$$

Solution (e)

$$f^*(y) = \sup_{x \in \mathbb{R}^n} (y^T x - \|x\|_1)$$

We can rewrite this as

$$y^T x - \|x\|_1 = \sum_{i=1}^n y_i x_i - \sum_{i=1}^n \|x_i\| = \sum_{i=1}^n (y_i x_i - \|x_i\|)$$

Case 1: $|y_i| > 1$. By choosing x_i to have the same sign as y_i and let x_i's magnitude go to ∞, we can show that

$$y^T x - \|x\|_1 = t(\|y_i\| - 1) \to \infty$$
$$\therefore f^*(y) = \infty$$

Case 2: $|y_i| \leq 1$

$$y^T x \leq \|y\|_\infty \|x\|_1 \quad \text{(Holder's inequality)}$$

So for all x

$$y^T x - \|x\|_1 \leq (\|y\|_\infty - 1)\|x\|_1$$

We have $\|y\|_\infty \leq 1$, so we can plug this into the inequality above, and get

$$y^T x - \|x\|_1 \leq 0 \quad \forall x$$

Conjugate function is

$$f^*(y) = \begin{cases} 0 & \|y\|_\infty \leq 1 \\ \infty & \text{otherwise} \end{cases}$$

Solution (f)

Let

$$f(x) = \frac{1}{x+1}, \qquad \operatorname{dom} f = \mathbb{R}_+.$$

The Fenchel conjugate of f is defined as

$$f^*(y) = \sup_{x \geq 0} \left(xy - \frac{1}{x+1} \right).$$

Define

$$g(x) = xy - \frac{1}{x+1}, \qquad x \geq 0.$$

Compute the derivative. We compute

$$g'(x) = y + \frac{1}{(x+1)^2}.$$

Setting $g'(x) = 0$ gives

$$y + \frac{1}{(x+1)^2} = 0 \quad \Longleftrightarrow \quad y = -\frac{1}{(x+1)^2}.$$

Thus, a critical point exists only if $y < 0$, and in this case

$$x + 1 = \sqrt{-\frac{1}{y}}, \qquad x = \sqrt{-\frac{1}{y}} - 1.$$

This point is feasible if and only if $x \geq 0$, which requires

$$\sqrt{-\frac{1}{y}} \geq 1 \quad \Longleftrightarrow \quad y \in [-1, 0).$$

Case analysis

Case 1: $y > 0$ Then $g'(x) > 0$ for all $x \geq 0$, so g is increasing and

$$\sup_{x \geq 0} g(x) = +\infty.$$

Hence $f^*(y) = +\infty$.

Case 2: $y = 0$**.** Then

$$g(x) = -\frac{1}{x+1} \leq 0,$$

and $\sup_{x \geq 0} g(x) = 0$ as $x \to \infty$. Thus,

$$f^*(0) = 0.$$

Case 3: $-1 \leq y < 0$**.** Then the critical point

$$x^* = \sqrt{-\frac{1}{y}} - 1$$

is feasible and maximizes g. Let $a = \sqrt{-1/y}$. Then $x^* = a - 1$ and $y = -1/a^2$. We compute

$$\begin{aligned} f^*(y) = g(x^*) &= y(a-1) - \frac{1}{a} \\ &= -\frac{1}{a^2}(a-1) - \frac{1}{a} \\ &= -\frac{2}{a} + \frac{1}{a^2}. \end{aligned}$$

Since $1/a = \sqrt{-y}$ and $1/a^2 = -y$, we obtain

$$f^*(y) = -y - 2\sqrt{-y}.$$

Case 4: $y < -1$**.** Then the critical point is infeasible. Moreover,

$$g'(0) = y + 1 < 0,$$

so g is decreasing on $[0, \infty)$. Hence the maximum occurs at $x = 0$:

$$f^*(y) = g(0) = -1.$$

Final expression. Combining all cases, we obtain

$$f^*(y) = \begin{cases} +\infty, & y > 0, \\ 0, & y = 0, \\ -y - 2\sqrt{-y}, & -1 \leq y < 0, \\ -1, & y < -1. \end{cases}$$

This is the Fenchel conjugate of $f(x) = \dfrac{1}{x+1}$ on $\mathbb{R}_+$